

When Should a Causal Model Refuse to Answer?

Paper discussion: Jesson, Mindermann, Shalit, Gal — NeurIPS 2020

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Part 1 — Understanding the paper

- ① The problem: CATE estimation at scale.
- ② The insight: non-overlap = covariate shift.
- ③ The tool: Bayesian neural networks.
- ④ The result: uncertainty beats propensity trimming.

Part 2 — Critical evaluation

- ① Assumption audit.
- ② What uncertainty does *not* fix.
- ③ The benchmarking problem.
- ④ Comparison to classical methods.
- ⑤ Discussion questions.

Story: a model makes a treatment recommendation

A hospital deploys a neural network to recommend cancer treatment.

- A new patient's profile — age, genetics, history — is unlike anyone in the training data.
- The model has seen no one like this under either treatment arm.
- The model outputs: **“Recommend treatment. Confidence: 94%.”**

The problem

Standard causal neural networks have no mechanism to say “I don't know.” They extrapolate silently and confidently into uncharted covariate space.

Course recap: CATE and identification

The individual-level estimand (from our notation):

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$$

Under **unconfoundedness** $(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$ and **consistency**:

$$\tau(x) = \underbrace{\mathbb{E}[Y \mid T = 1, X = x]}_{\mu_1(x)} - \underbrace{\mathbb{E}[Y \mid T = 0, X = x]}_{\mu_0(x)}$$

- $\mu_1(x)$ requires observed *treated* units near x .
- $\mu_0(x)$ requires observed *untreated* units near x .
- Both require **positivity**: $0 < e(x) < 1$.

Positivity fails silently in high dimensions

Positivity assumption

$$0 < \mathbb{P}(T = 1 \mid X = x) < 1 \quad \text{for all } x \text{ in the support of } X.$$

- In low dimensions: checked with propensity score histograms.
- In high dimensions (images, EHRs, genomics): every individual is almost unique — positivity fails somewhere for almost everyone.
- Consequence: some $\mu_1(x)$ or $\mu_0(x)$ must be **extrapolated**, not estimated.
- **Standard causal neural networks do not flag when they are extrapolating.**

Classical fix: propensity score trimming

- 1 Estimate $\hat{e}(x) = \hat{\mathbb{P}}(T = 1 \mid X = x)$.
- 2 **Trim:** remove units where $\hat{e}(x) < \eta$ or $\hat{e}(x) > 1 - \eta$.
- 3 Estimate CATE on the remaining data.

Intuition: units near $e(x) = 0$ or $e(x) = 1$ have few counterparts in the opposite arm — their CATE estimates are unreliable.

The fundamental limit of trimming

The propensity score captures *marginal* probability $\mathbb{P}(T = 1 \mid x)$. It does not directly tell you which arm lacks data for a specific x . As we will see, this can lead to trimming exactly the wrong units.

When trimming fails: the CEMNIST story

New benchmark introduced in the paper: **causal-effect MNIST**.

Digit(s)	$p(x)$	$e(x) = \mathbb{P}(T=1 \mid x)$	CATE	Estimable?
9	0.50	$1/9 \approx 0.11$	-1	Yes — many untreated nines
2	0.056	1.00	+1	No — only treated twos
other odd	0.056	0.50	-1	Yes
other even	0.056	0.50	+1	Yes

- Digit 9: low propensity (≈ 0.11), but plenty of untreated nines $\Rightarrow$ CATE *is* estimable.
- Digit 2: propensity = 1, no untreated twos $\Rightarrow$ CATE is *not* estimable.
- A trimmer with $\eta = 0.05$ keeps twos, removes nines — exactly backwards.

Core insight: non-overlap is covariate shift

Recall: covariate shift = the test distribution differs from the training distribution.

Consider estimating $\mu_1(x) = \mathbb{E}[Y \mid T = 1, X = x]$. The model for the $T = 1$ arm is trained on data from $p(X \mid T = 1)$.

The reframing (Jesson et al.)

If $e(x) = \mathbb{P}(T = 1 \mid x) \approx 0$, then by Bayes' theorem:

$$p(x \mid T = 1) \approx 0.$$

So x is **out-of-distribution** for the $T = 1$ model — a covariate shift.

- $e(x) \approx 0$: $\mu_1(x)$ is OoD.
- $e(x) \approx 1$: $\mu_0(x)$ is OoD.
- Non-overlap $\Leftrightarrow$ at least one outcome model must extrapolate.

Consequence: we can address both non-overlap *and* input covariate shift with a single tool 8/26

Epistemic vs. aleatoric uncertainty

Aleatoric uncertainty

Inherent noise in Y .

Example: a fair coin — even with unlimited data you cannot predict the flip.

Cannot be reduced with more observations.

Epistemic uncertainty

Uncertainty due to *lack of data*.

Example: no treated units like this patient
 $\Rightarrow$ uncertain what $\mu_1(x)$ is.

Disappears as more relevant data arrives.

We want to detect **epistemic** uncertainty in $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$,
and use it to decide when to withhold a recommendation.

Bayesian neural networks: the tool

Replace a single network $\mu_0(x; \omega_0)$ with a *distribution over networks*:

$$p(\omega_0 \mid \mathcal{D}) \quad (\text{posterior over weights}).$$

Epistemic uncertainty $\approx$ variance of predictions across sampled networks:

$$U_0(x) = \text{Var}_{\omega_0 \sim p(\omega_0 \mid \mathcal{D})} [\mu_0(x; \omega_0)].$$

- **In-distribution** x : all networks agree $\Rightarrow$ low variance.
- **OoD** x : networks disagree $\Rightarrow$ high variance.

Practical approximation: MC Dropout (Gal & Ghahramani, 2016)

Add dropout to each outcome head. Keep dropout *on* at test time. Each stochastic forward pass $\approx$ one draw from the approximate posterior.

CATE uncertainty: the formal decomposition

By the law of total variance:

$$\text{Var}[Y(1) - Y(0) \mid x] = \underbrace{\mathbb{E}_{\omega}[\text{Var}_{\mathcal{Y}}[Y(1) - Y(0) \mid \mu_{\omega_1}(x), \mu_{\omega_0}(x)]]}_{\text{aleatoric}} + \underbrace{\text{Var}_{\omega}[\mu_{\omega_1}(x) - \mu_{\omega_0}(x)]}_{\text{epistemic in } \hat{\tau}(x)}$$

The paper uses the epistemic term as its uncertainty signal:

$$U(x) = \text{Var}_{\omega}[\hat{\tau}(x)] = \text{Var}_{\omega}[\mu_{\omega_1}(x) - \mu_{\omega_0}(x)].$$

- Estimated by sampling ω via MC Dropout at test time.
- $U(x)$ is large when at least one arm model is OoD at x .

Neural CATE estimators: a brief map

Model	Key idea	Bayesian extension
T-Learner	Fit μ_0, μ_1 separately	B-TLearner
TARNet	Shared representation, separate outcome heads	BTARNet
CFRNet	TARNet + MMD regularisation for overlap	BCFR-MMD
Dragonnet	TARNet + joint propensity prediction head	BDragonnet
CEVAE	Latent confounders via VAE	BCEVAE + neg. sampling

Paper's recipe: add MC Dropout to the outcome heads of any model $\Rightarrow$ get $U(x)$ for free, with minimal code changes.

CEVAE special case: non-overlap inputs are generated by flipping $t \rightarrow 1 - t$ on training points (negative sampling) — tractable because the OoD distribution is known.

Rejection policy: thresholding on uncertainty

Decision rule: given a rejection budget r_{rej} , withhold recommendation for the r_{rej} fraction of test units with highest $U(x)$.

Uncertainty policy

reject x if $U(x) > \kappa$

- Model-based; detects non-overlap and covariate shift.
- Threshold κ set to achieve target r_{rej} .

Propensity trimming

reject x if $\hat{e}(x) < \eta$ or $> 1-\eta$

- Only captures marginal probability.
- Trims low-e units even when their CATE is estimable (digit 9).

Evaluation metric: recommendation error rate and $\sqrt{\text{PEHE}}$ on non-rejected subset, at fixed r_{rej} .

Results: CEMNIST ($r_{\text{rej}} = 0.50$)

Model	Random	Propensity	Uncertainty
BART	2.1	2.1	2.0
B-TLearner	0.3	0.2	0.0
BTARNet	0.2	0.2	0.0
BCFR-MMD	0.3	0.3	0.1
BDragonnet	0.2	0.2	0.0
BCEVAE	0.3	0.2	0.0

$\sqrt{\text{PEHE}}$, lower is better.

- Uncertainty policy dominates across all architectures.
- Propensity policy trims nines (estimable!) instead of twos (not estimable). Uncertainty correctly identifies where the model lacks data.

Results: IHDP covariate shift and ACIC 2016

Setup: train on IHDP excluding unmarried mothers; test includes them.

Model	IHDP Cov. Shift ($r_{\text{rej}} = 0.50$)			ACIC 2016 ($r_{\text{rej}} = 0.10$)		
	Rand.	Prop.	Unct.	Rand.	Prop.	Unct.
BART	2.6	2.7	1.8	1.3	1.2	0.9
BTARNet	2.2	2.0	1.2	1.8	1.7	1.2
BDragonnet	2.4	2.2	1.3	1.9	1.8	1.3

$\sqrt{\text{PEHE}}$, lower is better.

- Uncertainty is more data-efficient: lower error at the same rejection rate.
- Result consistent across all model classes and all three benchmarks.

What Jesson et al. (2020) show

- ① Non-overlap = covariate shift for one outcome model.
- ② Epistemic uncertainty in BNNs detects this — and detects input covariate shift as a bonus.
- ③ MC Dropout is a cheap, practical approximation.
- ④ Uncertainty-based rejection outperforms propensity trimming, especially when $e(x)$ and CATE estimability are decorrelated.

Now: how much should we trust this?

Assumption audit

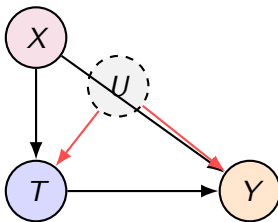
Assumption	Paper status	Notes
Unconfoundedness: $(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$	Assumed, untested	Foundation of all CATE identification
SUTVA / consistency	Assumed, untested	Standard in this literature
Positivity $0 < e(x) < 1$	Relaxed	The paper's contribution
Correct model class	Assumed	NN may misspecify $\mu_t(x)$
i.i.d. training data	Assumed	Violated under deployment shift

The unconfoundedness blindspot

The paper's method tells you **when the model is uncertain**.
It does not tell you **whether the model is biased**.

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$$

If unconfoundedness fails — a hidden U confounds both T and Y :



- $\hat{\mu}_1(x)$ and $\hat{\mu}_0(x)$ absorb the confounding signal.
- Their difference $\hat{\tau}(x)$ is biased.
- $U(x) = \text{Var}_\omega[\hat{\tau}(x)]$ can be small even when bias is large.

Two failure modes: only one is addressed

	Low bias	High bias (hidden confounding)
High $U(x)$	Reject $\Rightarrow$ safe. Paper detects this.	Reject $\Rightarrow$ safe (by luck). Paper detects this too, but for the wrong reason.
Low $U(x)$	Recommend $\Rightarrow$ safe.	Recommend $\Rightarrow$ confidently wrong. The paper's blind spot.

A Bayesian causal model can be *confident and wrong* if the data is confounded — not just if overlap is violated.

What about sensitivity analysis?

Classical causal inference addresses hidden confounding directly:

- **Rosenbaum bounds:** how large must a hidden U be to overturn the conclusion?
- **E-values:** minimum association strength of a confounder to explain away the estimated effect.
- **Random common cause refuters** (DoWhy): add a synthetic unobserved confounder; check if the estimate changes.

The paper has no sensitivity analysis.

It tells us *when* to trust a CATE estimate (in-distribution, overlap region). It does not tell us *how robust* the estimate is to hidden confounding.

These are **orthogonal questions**. Uncertainty quantification and sensitivity analysis both belong in a complete causal inference pipeline. This paper provides one, not both.

The benchmarking problem

All experiments use **semi-synthetic** datasets:

- **IHDP**: real covariates from a clinical trial; synthetic outcome model.
- **ACIC 2016**: synthetic data from a methods competition; known DGP.
- **CEMNIST**: MNIST digits; synthetic treatment and outcome assignment.

In semi-synthetic data, the true CATE is **known by construction** — that is why $\sqrt{\text{PEHE}}$ can be computed.

In real-world observational data

$Y_i(0)$ for treated units is **never observed**.

$\sqrt{\text{PEHE}}$ is **uncomputable**.

We cannot verify the paper's claims in deployment.

This is not a flaw unique to this paper — it is the fundamental evaluation problem in all individual-level causal effect estimation.

Comparison to classical methods

	Propensity trimming	Uncertainty rejection
Detects non-overlap	Yes, via $\hat{e}(x)$	Yes, more precisely
Handles covariate shift	No	Yes
Handles hidden confounding	No	No
Sensitivity analysis	Via Rosenbaum bounds / E-values	None
Computationally cheap	Yes	No — requires BNN training
Principled threshold	Yes (Caliendo & Kopeinig)	No principled rule given
Verifiable on real data	Yes (no CATE ground truth needed)	Yes, but performance unverifiable

What the paper genuinely contributes

- ① **Elegant reframing.** Non-overlap = covariate shift for one arm. This is clean, theoretically well-motivated, and opens a new toolkit.
- ② **Practical method.** MC Dropout adds uncertainty to any neural causal model with minimal engineering cost.
- ③ **New benchmark.** CEMNIST exposes a concrete, reproducible failure mode of propensity-based trimming in high-dimensional data.
- ④ **Better rejection.** Outperforms propensity trimming consistently across three benchmarks and five model architectures.

What it does not contribute

A solution to hidden confounding, a sensitivity analysis, or any mechanism to validate CATE estimates on real-world data where ground truth is unavailable.

Discussion questions — Part A

- 1 **Uncertainty and confounding.** The paper assumes unconfoundedness throughout. If there are hidden confounders, can high epistemic uncertainty $U(x)$ protect you? What would need to change in the framework to also catch confounding bias, not just data sparsity?
- 2 **Real-world validation.** The paper shows uncertainty beats trimming on IHDP and CEMNIST — both semi-synthetic. What experiment, using only observational data with no known ground truth, would convince you this result holds in a clinical deployment?
- 3 **Redundancy with CFRNet.** CFRNet already penalises for overlap violations via MMD regularisation. Does adding MC Dropout on top give something genuinely new, or is it partially redundant? What does each mechanism address, precisely?

- ④ **What has the model contributed?** Suppose you run the paper's method on a new patient and get very high $U(x)$. You withhold the recommendation and defer to a clinician. What has the causal model actually contributed to the decision? Is detecting uncertainty enough to be useful? What else would you need?
- ⑤ **Uncertainty about what, exactly?** The paper estimates $U(x) = \text{Var}_\omega[\hat{\tau}(x)]$ — variance across BNN draws. Is this the same as uncertainty about the true causal effect $Y(1) - Y(0)$? What components of total uncertainty are missing from $U(x)$? (Think: model misspecification, hidden confounders, estimation noise.)

$$U(x) = \text{Var}_\omega [\hat{\mu}_{\omega_1}(x) - \hat{\mu}_{\omega_0}(x)]$$

- **What the paper gives you:** a principled, model-based signal for when to withhold a CATE estimate, detecting non-overlap and covariate shift simultaneously.
- **What it assumes and does not question:** unconfoundedness, correct model class, i.i.d. data, and SUTVA. High $U(x)$ is not a guard against confounding bias — a model can be uncertain for the right reason and still be wrong for a different reason.

The deeper lesson for this course

Identification and uncertainty quantification are orthogonal.

You can know exactly how uncertain you are about the wrong estimand.

Epistemic uncertainty $\neq$ robustness to assumption violations.